

Tutorial Sheet 5

Pontryagin's Minimum Principle

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x

Exercise 1. Consider a dynamical system

$$\dot{x} = f(x) + g(x)u,$$

and suppose we wish to design the control input u to minimise a cost functional of the form

$$J(u, x(0)) = \frac{1}{2} \int_0^\infty q(x) + u^\top R u dt,$$

where $q(x) \geq 0$, $q(0) = 0$ and $R = R^\top > 0$.

- Write the state-costate dynamics associated with the optimal control problem.
- What is the Hamiltonian function associated with the problem.
- Show that the value of the Hamiltonian function is constant.

$$a) \quad H(x, u, \lambda) = q(x) + u^\top R u + \lambda^\top (f + gu)$$

Therefore, the state-costate dynamics are given by

$$\dot{x} = \left(\frac{\partial H}{\partial \lambda} \right)^\top = \boxed{f^\top + u^\top g^\top} \quad \frac{1}{2} (f(x) + g(x)u)$$

$$\dot{\lambda} = - \left(\frac{\partial H}{\partial x} \right)^\top = - \left(\frac{dq}{dx} \right)^\top - \frac{1}{2} \left(\frac{df}{dx} + \boxed{\frac{dg}{du}} \right)^\top \lambda \quad \frac{\partial q}{\partial x} u$$

$$b) \quad H(x, u, \lambda) = \frac{1}{2} q(x) + \frac{1}{2} u^\top R u + \lambda^\top (f + gu)$$

$$c) \quad \frac{\partial H}{\partial t} = \frac{\partial H}{\partial x} \frac{\partial x}{\partial t} + \frac{\partial H}{\partial u} \frac{\partial u}{\partial t} + \frac{\partial H}{\partial \lambda} \frac{\partial \lambda}{\partial t}$$

$$= -\dot{\lambda} \cdot \dot{x} + 0 \cdot \dot{u} + \dot{x} \cdot \lambda \quad \text{for } u = u^*$$

$$= 0$$

$\therefore$ the value of Hamiltonian function is constant

Exercise 2. Consider a dynamical system

$$\dot{x} = f(x) + g(x)u,$$

and suppose we wish to design the control input u to minimise a cost functional of the form

$$J(u, x(0)) = \int_0^\infty q(x) + u^\top R u dt,$$

where $q(x) \geq 0$, $q(0) = 0$ and $R = R^\top > 0$. You can assume that all functions associated with the optimal control problem are well-defined and sufficiently smooth.

- a) What are the state-costate dynamics associated with the optimal control problem?
- b) What is the Hamilton-Jacobi-Bellman equation associated with the problem?
- c) A manifold is essentially a type of subset of the state space (so, in our case, a type of subset of $\mathbb{R}^n$). A manifold is said to be *invariant* if it satisfies the property that if the state is on the manifold at a given time, it remains there for all time.

Show that

$$\lambda = \frac{\partial V^\top}{\partial x},$$

where $\frac{\partial V}{\partial x}$ is a row vector containing the partial derivatives of $V(x)$ with respect to the components of x , is an invariant manifold.

Hint: Define a new variable $z = \lambda - \frac{\partial V}{\partial x}^\top$ and show that if $z(0) = 0$, then $z(t) = 0$ for all time $t \geq 0$.

$$a) \quad \dot{x} = \left(\frac{\partial H}{\partial \lambda}\right)^\top \quad \text{and} \quad \dot{\lambda} = -\left(\frac{\partial H}{\partial x}\right)^\top$$

$$b) \quad H(x, u, \lambda) = \underbrace{\frac{1}{2}}_{\lambda} q(x) + \underbrace{\frac{1}{2}}_{\lambda} u^\top R u + \underbrace{(\nabla V)^\top}_{\lambda} [f(x) + g(x, u)] \quad * \quad (\nabla V)^\top = \frac{\partial V}{\partial x}$$

$$c) \quad \text{let } z = \lambda - \left(\frac{\partial V}{\partial x}\right)^\top \quad \text{and} \quad z(0) = 0$$

$$\frac{\partial z}{\partial t} = \frac{\partial \lambda}{\partial t} - \frac{\partial}{\partial t} \left(\frac{\partial V}{\partial x}\right)^\top$$

$$= -\frac{\partial H}{\partial x} - \frac{\partial}{\partial t} \left(\frac{\partial V}{\partial x}\right)^\top$$

$$= -\frac{\partial}{\partial x} \left(H + \frac{\partial V}{\partial t} \right)$$

$$= -\frac{\partial}{\partial x} (H - H)$$

$$= 0$$

$$\therefore z(t) = \int \frac{\partial 0}{\partial t} dt = 0$$

Exercise 3. Consider the optimal control problem defined by the dynamics

$$\dot{x} = Ax + Bu,$$

and the cost functional

$$J(u, x_0) = \frac{1}{2} \int_0^T x(t)^\top Q x(t) + u(t)^\top R u(t) dt + \frac{1}{2} x(T)^\top Q_T x(T).$$

The initial condition of the state is $x(0) = x_0$.

- a) Show that the state-costate dynamics corresponding to this optimal control problem are of the form

$$\begin{bmatrix} \dot{x} \\ \dot{\lambda} \end{bmatrix} = H \begin{bmatrix} x \\ \lambda \end{bmatrix},$$

and provide an expression for the matrix H in terms of the dynamics and cost matrices.

- b) Write the differential Riccati equation associated with the optimal control problem.
- c) Let $z(t) = \lambda(t) - P(t)x(t)$, where $P(t)$ is the solution of the differential Riccati equation in Part (b).

(i) Derive an expression for the dynamics of $z(t)$.

(ii) Show that if $z(0) = 0$, then $z(t) = 0$, for all $t \geq 0$.

- d) Let

$$A = \begin{bmatrix} -1 & 0.1 \\ 0.5 & -1 \end{bmatrix}, \quad B = \begin{bmatrix} -0.1 \\ 0.1 \end{bmatrix}$$

$Q = I$, $Q_T = 0$, and $R = I$.

(i) Compute the eigenvalues of the matrix H for the above values of the dynamics and cost matrices.

(ii) Comment on the stability properties of the state/costate system.

- (iii) State the boundary conditions of the state and costate that correspond to the optimal state/costate trajectory.
- (iv) Solve the two-point boundary value problem that yields the optimal state/costate trajectory corresponding to the initial condition of the state being $x_0 = [5, 5]^T$ numerically. You may find it useful to use an inbuilt function such as Matlab's `bvp4c`.
- (v) What is the optimal control, in terms of the optimal costate trajectory?
- (vi) An inbuilt function such as `bvp4c` uses efficient ode solvers (and variable time steps). Thus, you will likely find that it performs well in many settings. Nevertheless, explaining what issues you expect will arise as T becomes large?
- (vii) Suppose now that the initial condition x_0 is different. Can you "re-use" the result computed for x_0 given in Part (iv) of this question directly to construct the optimal control?

$$a) \quad H(x, u, \lambda) = \frac{1}{2} x^T Q x + \frac{1}{2} u^T R u + \lambda^T (A x + B u)$$

$$\dot{x} = \left(\frac{\partial H}{\partial \lambda} \right)^T = A x + B u$$

$$\dot{\lambda} = \left(-\frac{\partial H}{\partial x} \right)^T = -Q x - A^T \lambda$$

$$\left. \frac{\partial H}{\partial u} \right|_{u=u^*} = u^T R + \lambda^T B = 0$$

$$\therefore u = u^* = -R^{-1} (\lambda^T B)^T = -R^{-1} B^T \lambda$$

$$\therefore \dot{x} = A x - B R^{-1} B^T \lambda$$

$$\therefore \begin{bmatrix} \dot{x} \\ \dot{\lambda} \end{bmatrix} = \begin{bmatrix} A & -B R^{-1} B^T \\ -Q & -A^T \end{bmatrix} \begin{bmatrix} x \\ \lambda \end{bmatrix}$$

$$b) \quad \text{let } \lambda = P(t) x(t), \text{ therefore}$$

$$\dot{\lambda} = \dot{P}x + P\dot{x} = \dot{P}x + PAx - PBR^{-1}B^T Px$$

$$= -Qx - A^T Px$$

$$\therefore -\dot{P} = A^T P + PA - PBR^{-1}B^T P + Q$$

$$c) (i) \dot{z} = \frac{d}{dt}(\lambda - Px)$$

$$= \dot{\lambda} - \dot{P}x - P\dot{x}$$

$$= -Qx - A^T \lambda - \dot{P}x - PAx + \overset{P}{\wedge} BR^{-1}B^T \lambda$$

$$= (-A^T + \overset{P}{\wedge} BR^{-1}B^T) \lambda + (A^T P - PBR^{-1}B^T P) x$$

$$= (-A^T + \overset{P}{\wedge} BR^{-1}B^T)(\lambda - Px)$$

$$= (-A^T + \overset{P}{\wedge} BR^{-1}B^T) z$$

$$(ii) \ln(z) = (-A^T - BR^{-1}B^T)t + K$$

$$\therefore z = Ce^{-(A^T + R^{-1}B^T)t}$$

$$\therefore z(0) = Ce = 0 \quad \therefore C = 0$$

$$\therefore z = 0$$

when $z=0$,

$$\dot{z} = z = 0$$

$\therefore$ system is at the equilibrium point,
will not change over the time

Therefore, $z(t) = 0$

d) (i)

$$\lambda_H = \begin{bmatrix} -1.2324 \\ -0.7754 \\ 0.7754 \\ 1.2324 \end{bmatrix}$$

(ii) the system is ^{un}stable since P correspond to stable subspace only

diii) $x(0) = x_0$ and $\lambda(T) = \left(\frac{\partial \Phi}{\partial x} \Big|_{x=x(T)} \right)^T = \frac{1}{2} [x(T)^T Q_T]^T = \frac{1}{2} Q_T x(T)$

iv) v) see MATLAB, $u^* = -R^{-1} B^T \lambda$

vi) numerical issue arise, the PMP become numerical unstable,

this is because state/costate dynamic is unstable, part of the open-loop system will growth exponentially over time when guess the $\lambda(0)$

vii) ~~If $P(t)$ has been calculated, then yes~~

No, since PMP only yield open-loop solution